



AURA Data Layer — “DTATA” (Data Spine Report)

1. Overview

The **DTATA layer** represents the unified data backbone of AURA (Adaptive Unified Real Estate Intelligence Engine).

Instead of relying on a single dataset, DTATA is constructed through **multi-source data fusion**, combining:

- market listings (supply & sentiment)
- official statistics (ground truth)
- macroeconomic signals (demand drivers)
- ESG indicators (future value)
- regulatory context (risk layer)

This architecture ensures that AURA remains:

- **decision-oriented**
 - **explainable**
 - **aligned with real-world market dynamics**
-



2. DTATA Core Dataset Components



CATEGORY 1 — Property Listings (Market Layer)

- **Dataset Name:** Portugal Real Estate Listings 2024–2025
- **Source:** Kaggle
- **Direct Link:** <https://www.kaggle.com/datasets/luvathoms/portugal-real-estate-2024>
- **Access Type:** CSV

Description

A large-scale dataset (100,000+ properties) aggregated from major Portuguese real estate platforms such as Idealista and Imovirtual. It includes detailed property-level features.

Key Features

- price
- area (multiple types)
- rooms
- location (city, district)
- property type
- energy rating (partial)

Role in AURA

👉 Serves as the **primary market layer**, capturing:

- seller pricing behavior
- inventory dynamics
- feature-level signals

● CATEGORY 1.2 — Listing Calibration & Market Dynamics Layer

- **Dataset Name:** Portugal Housing Affordability and Market Factors
- **Source:** Kaggle (Integrated Multi-Source Dataset)
- **Direct Link:** <https://www.kaggle.com/datasets/marquesigor/portugal-housing-affordability-and-market-factors>
- **Access Type:** CSV / Time-Series Data



Description

This dataset provides a **longitudinal, multi-source view of the Portuguese housing market**, combining:

- historical listing prices (€/m²)
- transaction-level indicators
- affordability metrics
- regional market evolution

It spans from **2009 to 2025**, offering a **time-aware perspective** that complements the static listing dataset.



Granularity

- Regional (NUTS / districts depending on source aggregation)
 - Time-based (quarterly / yearly trends)
-



Role in AURA

👉 Serves as the **calibration and temporal intelligence layer** for Category 1 listings.

While Category 1.1 captures **current market supply and seller behavior**, this dataset introduces:



1. Time-Aware Market Context

- tracks long-term price evolution
 - identifies cycles (growth, stagnation, correction)
-



2. Asking vs Reality Alignment

- bridges the gap between:
 - listing prices (often inflated)
 - real market conditions

👉 Enables AURA to detect:

- overpricing
 - underpricing
 - realistic valuation bands
-

3. Demand & Affordability Signals

- integrates affordability metrics
- reflects buyer purchasing power

 Critical for:

- liquidity estimation
 - demand pressure modeling
-

4. Market Momentum Features

Enables creation of advanced features such as:

- PriceTrend_12M
 - MarketMomentumScore
 - AffordabilityIndex
 - DemandPressureScore
-

Integration with Category 1.1

This dataset must be **joined with the primary listings dataset** using:

- **location alignment** (district / region)
 - **time alignment** (month / quarter / year)
-

Example Use in AURA Pipeline

- normalize listing prices against historical averages
- detect if current prices are above/below trend
- adjust valuation predictions based on macro housing cycles

- enhance seller decision logic (SELL / WAIT / REDUCE_PRICE)
-

Strategic Importance

Without this dataset:

✗ AURA behaves like a static pricing model

With this dataset:

✓ AURA becomes a **dynamic, time-aware market intelligence system**

CATEGORY 2 — Official Market Data (Ground Truth Layer)

- **Dataset Name:** Housing Price Index (HPI) – Indicator 0009201
- **Source:** Statistics Portugal
- **Direct Link:** https://www.ine.pt/xportal/xmain?xpid=INE&xpgid=ine_indicadores&indOcorrCod=0009201
- **Access Type:** API / CSV

Description

Official quarterly index tracking price evolution of residential properties (new and existing).

Granularity

- National
- NUTS II / NUTS III

Role in AURA

👉 Acts as the **market calibration anchor**:

- normalizes listing prices
- distinguishes real appreciation vs inflation

- ensures model outputs reflect actual transactions
-

🟡 CATEGORY 3 — Financial & Macro Data (Demand Layer)

- **Dataset Name:** BPstat Housing Credit & Interest Rates
- **Source:** Banco de Portugal
- **Direct Link:** <https://bpstat.bportugal.pt/>
- **Access Type:** API / CSV

Description

Real-time financial indicators including:

- mortgage interest rates
- Euribor
- housing credit conditions

Granularity

- National

Role in AURA

👉 Powers the **demand and liquidity engine**:

- models buyer affordability
 - simulates interest-rate shocks
 - explains demand-side pressure
-

🟣 CATEGORY 4 — Socioeconomic Context (Neighborhood Intelligence)

- **Dataset Name:** PORDATA Portugal Database
- **Source:** PORDATA

- **Direct Link:** <https://www.pordata.pt/en/portugal>
- **Access Type:** CSV / Excel

Description

Comprehensive dataset covering:

- population
- migration
- income
- labor statistics

Granularity

- Municipality / Region

Role in AURA

👉 Enables **neighborhood strength modeling**:

- links property value to demographics
- captures migration-driven demand
- enhances explainability

● CATEGORY 5 — ESG / Energy Intelligence (Sustainability Layer)

● CATEGORY 5.1 — Energy Certification Data (Unavailable)

- **Dataset Name:** SCE Energy Certification Database
- **Source:** ADENE
- **Direct Link:** <https://dados.gov.pt/pt/datasets/sce-sistema-de-certificacao-energetica-de-edificios/>
- **Access Type:** Portal / CSV

Description

National database of building energy performance certificates (A+ to F), representing official energy efficiency ratings assigned to properties.

Granularity

- Property level
- Parish level

Role in AURA

👉 Intended to power the **Green Value Engine**:

- estimate renovation ROI
- capture energy efficiency premium
- integrate ESG into valuation

⚠️ Limitation

This dataset was not accessible in a usable format during the development phase. Therefore, direct EPC-based energy labeling could not be integrated into the AURA prototype.

🟢 CATEGORY 5.2 — EUBUCCO (Attempted Source — Not Used)

1. EUBUCCO (European Building Stock Characteristics)

- **Source:** EUBUCCO Consortium
- **Direct Link:** <https://eubucco.com/data/>
- **Coverage:** Portugal (EU-wide dataset)

Description

A large-scale European dataset containing building-level characteristics such as geometry, height, floors, construction year, and usage type.

Role in AURA

- **EnergyScore:** estimation via building age and structure
- **GreenPremium:** volume-based efficiency proxy

Limitation

Although highly relevant, the dataset was not practically usable due to access constraints. The official source was unavailable, and alternative access via Zenodo required downloading extremely large archives (~100GB) to retrieve a small Portugal subset (~7MB), making it impractical for this project.

CATEGORY 5.3 — Building Footprint Dataset (Final Adopted Solution)

- **Dataset Name:** Portugal Buildings (OpenStreetMap Export)
- **Source:** Humanitarian OpenStreetMap Team
- **Direct Link:** https://data.humdata.org/dataset/hotosm_prt_buildings
- **Access Type:** GeoJSON / GIS

Description

A nationwide dataset containing building footprints extracted from OpenStreetMap, covering over 2 million buildings across Portugal. Each record represents a building polygon with geographic and structural attributes.

Granularity

- Building-level
- Spatial (geometry-based)

Key Variables

- Geometry (building footprint)
- Location (latitude / longitude)

- Building type (residential, house, etc.)
-

Role in AURA

👉 Powers the **ESG Proxy Engine**:

🔥 EnergyScore (proxy-based)

- footprint → surface exposure
- density → urban heat / efficiency

🔥 GreenPremium

- spatial distribution
- urban structure quality

🔥 RenovationPotential

- indirect estimation via density and typology
-

💡 Justification

This dataset was selected as a practical and scalable alternative to unavailable EPC and EUBUCCO data. While it does not provide direct energy performance indicators, it enables the construction of proxy-based ESG features through spatial and structural analysis.

⚠️ Limitations

- No direct energy ratings (A+ to F)
 - No building age or construction year
 - No thermal or material properties
 - Requires integration with external models (e.g., TABULA) for full energy estimation
-

CATEGORY 6 — Regulatory & Risk Context (Compliance Layer)

- **Dataset Name:** RNAL — Alojamento Local Registry
- **Source:** Turismo de Portugal
- **Direct Link:**
https://webservices.turismodeportugal.pt/RNT_External/rest/RNT/list_RNAL
- **Access Type:** API (JSON)

Description

Official national registry of short-term rental properties (Alojamento Local), including:

- property identification (NrRegisto)
- accommodation type (apartment, hostel, etc.)
- capacity (beds, guests)
- full address (street, postal code, parish)
- ownership information

Granularity

- Property-level (high resolution)

Role in AURA

 Enables **Regulatory & Demand Pressure Intelligence:**

- detects short-term rental density
- identifies tourism-heavy zones
- flags regulatory pressure areas
- enhances investor risk scoring

Limitation

The API endpoint exhibited instability (502 errors), but direct access returned valid JSON data. Extraction required manual handling and processing.

3. DTATA Architecture Summary

DTATA = Listings

- + Official Market Data
- + Macro & Financial Data
- + Socioeconomic Context
- + ESG / Energy Data
- + Regulatory Signals

4. Strategic Value for AURA

The DTATA layer transforms AURA from:

✗ a simple prediction model

➡ into

✓ a real estate intelligence system

Key Advantages

✓ Multi-Source Fusion

Avoids bias from single datasets

✓ Explainability

Each prediction is supported by:

- market data
- macro context
- risk indicators

✓ Decision-Oriented Output

Supports seller decisions such as:

- SELL_NOW
- WAIT
- REDUCE_PRICE
- HOLD

✓ Supervisor Alignment

Fully aligned with requirements:

- use all relevant data
- include external factors
- remain explainable
- focus on decisions

5. Data Integration Pipeline

The AURA system integrates multiple datasets through a structured pipeline:

Listings (Category 1)

↓

Calibrated with Official Price Index (Category 2)

↓

Adjusted by Financial & Macro Conditions (Category 3)

↓

Enriched with Socioeconomic Context (Category 4)

↓

Enhanced with ESG Proxies (Category 5)

↓

Adjusted for Regulatory Risk (Category 6)

↓

Final AURA Intelligence Layer

6. AURA Output Features

The integrated DATA layer enables AURA to generate decision-oriented outputs:

Core Outputs

- **Estimated Property Value**
- **Investment Score**
- **Demand Score**
- **Risk Score**
- **ESG Score**

Decision Engine Outputs

- SELL_NOW
- WAIT
- REDUCE_PRICE
- HOLD

Explanation Layer

Each output is supported by:

- market data signals
- macroeconomic conditions
- demographic indicators
- ESG proxies
- regulatory constraints

This ensures that AURA remains fully explainable and aligned with real-world decision-making.

7. Conclusion

The DTATA layer provides a **production-ready data foundation** for AURA.

By combining high-quality datasets across multiple dimensions, it ensures:

- robust modeling
- realistic valuation
- actionable insights

This approach positions AURA not as a static model, but as a **dynamic, adaptive intelligence engine** capable of evolving with the Portuguese real estate market.
